Problem 5

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Problem 1 *Let f and g be two functions such that $\lim_{x\to\infty} f(x) = \lim_{x\to\infty} g(x) = \infty$.*

*We say that f **grows faster** than g as x goes to ∞ if $\lim_{x\to\infty} \frac{f(x)}{g(x)} = \infty$, or, equivalently, if*

$\lim_{x\to\infty} \frac{g(x)}{f(x)} = 0.$

If the limit exists, namely, if $\lim_{x\to\infty} \frac{f(x)}{g(x)} = M$, for some positive number M,

*then we say that the functions f and g have **comparable growth rates**.*

In other words, we compare the growth rates of functions f and g by computing the limit of the form $\frac{\infty}{\infty}$: $\lim_{x\to\infty} \frac{f(x)}{g(x)}$.

For each of the following pairs of functions, determine which of the pair grows faster or state that they have comparable growth rates. Justify your answer by computing the limit $\lim_{x\to\infty} \frac{f(x)}{g(x)}$.

(a) $b^x; x^x; b > 1$

Explanation. $\lim_{x\to\infty} \frac{b^x}{x^x} = \lim_{x\to\infty} \left(\frac{b}{x}\right)^x = 0$. Since this limit is of the form 0^∞, which is a determinate form. Therefore, x^x grows faster.

(b) $x^x; \left(\frac{x}{e}\right)^x$

Explanation. $\lim_{x\to\infty} \frac{x^x}{(x/e)^x} = \lim_{x\to\infty} \frac{x^x}{\frac{x^x}{e^x}} = \lim_{x\to\infty} x^x \cdot \frac{e^x}{x^x} = \lim_{x\to\infty} e^x = \infty$ Therefore, x^x grows faster.

(c) $x^3; x^3 \cdot \ln(x)$

Explanation. $\lim_{x\to\infty} \frac{x^3}{x^3 \cdot \ln(x)} = \lim_{x\to\infty} \frac{1}{\ln(x)} = 0$. Therefore, $x^3 \cdot \ln(x)$ grows faster.

(d) $a^x; b^x; 0 < a < b$

Explanation. $\lim_{x\to\infty} \frac{a^x}{b^x} = \lim_{x\to\infty} \left(\frac{a}{b}\right)^x = 0$ Therefore, b^x grows faster

(e) $\log_a(x); \log_b(x); 1 < a < b$

Explanation. $\lim_{x\to\infty} \frac{\log_a(x)}{\log_b(x)}$

By L'Hopital's Rule: $\lim_{x\to\infty} \frac{\frac{1}{x\ln(a)}}{\frac{1}{x\ln(b)}} = \lim_{x\to\infty} \left(\frac{1}{x\ln(a)} \cdot \frac{x\ln(b)}{1}\right) = \frac{\ln b}{\ln a}$

Therefore, $\log_a(x)$ and $\log_b(x)$ grow at comparable rates.

(f) $\ln^3(x); x^{1/2}$

Explanation. $\lim_{x\to\infty} \frac{\ln^3(x)}{x^{1/2}}$

By L'Hopital's Rule: $\lim_{x\to\infty} \frac{\ln^3(x)}{x^{1/2}} = \lim_{x\to\infty} \frac{3\ln^2(x)\cdot(1/x)}{(1/2)x^{-1/2}} = \lim_{x\to\infty} \frac{6\cdot\ln^2(x)}{x^{1/2}}$

By L'Hopital's Rule: $\lim_{x\to\infty} \frac{12\ln(x)\cdot(1/x)}{(1/2)x^{-1/2}} = \lim_{x\to\infty} \frac{24\cdot\ln(x)}{x^{1/2}}$

By L'Hopital's Rule: $\lim_{x\to\infty} \frac{48}{x^{1/2}} = 0$ Therefore, $x^{1/2}$ grows faster

(g) $x; \ln(x)\sqrt{x}$

Explanation. $\lim_{x\to\infty} \frac{x}{\ln(x)\sqrt{x}} = \lim_{x\to\infty} \frac{x^{1/2}}{\ln(x)}$

By L'Hopital's Rule: $\lim_{x\to\infty} \frac{(1/2)x^{-1/2}}{x^{-1}} = \lim_{x\to\infty} (1/2)x^{1/2} = \infty$

Therefore, x grows faster

(h) *Challenge:* $x^{40}; 1.004^x$ *(Hint: Use the substitution* $x = \ln(t)$*.)*

Explanation.

$$
\begin{aligned}
\lim_{x\to\infty} \frac{x^{40}}{1.004^x} &= \lim_{t\to\infty} \frac{[\ln(t)]^{40}}{1.004^{\ln(t)}} \\
&= \lim_{t\to\infty} \frac{\ln(t)^{40}}{e^{\ln(1.004)^{\ln(t)}}} \\
&= \lim_{t\to\infty} \frac{\ln(t)^{40}}{t^{\ln(1.004)}} \\
&= \lim_{t\to\infty} \left(\frac{\ln(t)}{t^{1.004/40}}\right)^{40} \\
&= \left[\lim_{t\to\infty} \left(\frac{\ln(t)}{t^{1.004/40}}\right)\right]^{40} \\
&\overset{L.R.}{=} \left[\lim_{t\to\infty} \frac{\frac{1}{t}}{(1.004/40)t^{(1.004/40)-1}}\right]^{40} \\
&= \left[\lim_{t\to\infty} \frac{1}{(1.004/40)t^{((1.004/40)-1+1)}}\right]^{40} \\
&= 0^{40} \\
&= 0
\end{aligned}
$$

Therefore, 1.004^x grows faster.

(i) *Put the following functions in order of growth rate.*

$$x^3 \cdot \ln(x), \ln^3(x), x^x, 1.004^x, x^{40}, x^3$$

Explanation.

$$\ln^3(x) << x^3 << x^3 \cdot \ln(x) << x^{40} << 1.004^x << x^x$$